Math 54 — Week 2

Vector equations; the matrix equation $Ax = b$; solution sets (Lay 1.3–1.5)

§1 Facts used from Week 1

(W1) Row operations do not change the solution set of a linear system, and each row operation can be undone by another row operation.

(W2) A system is consistent exactly when an echelon form of its augmented matrix has no row $[\,0 \;\cdots\; 0 \;\; b\,]$ with $b \neq 0$. A consistent system has exactly one solution if there are no free variables, infinitely many otherwise.

(W3) Basic variables correspond to pivot columns; the others are free. Each pivot lies in a different row and a different column, so the number of pivots is at most the number of rows and at most the number of columns.

(W4) The pivot positions are the positions of the leading entries of any echelon form. In an echelon form every nonzero row has a leading entry, and the zero rows are at the bottom.

§2 This week

§2.1 Vectors and linear combinations (Lay 1.3)

The question. The system $x_1 + 2x_2 = 5$, $3x_1 + x_2 = 5$ asks for two numbers. Is there a single statement about the columns $(1, 3)$ and $(2, 1)$ of its coefficients that says the same thing as the two equations?

Definition. A **vector** in $\mathbb{R}^n$ is a column of n real numbers,

$$\mathbf{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}.$$

Two vectors are **equal** when all corresponding entries are equal. The **sum** $\mathbf{u} + \mathbf{v}$ has entries $u_i + v_i$; the **scalar multiple** $c\mathbf{u}$ has entries cu_i. The **zero vector** $\mathbf{0}$ has all entries 0. We write $-\mathbf{u}$ for $(-1)\mathbf{u}$ and $\mathbf{u} - \mathbf{v}$ for $\mathbf{u} + (-1)\mathbf{v}$.

- **Notation.** In running text the column vector is written $(u_1, \ldots, u_n)$, with commas. This is not the $1 \times n$ matrix $[\,u_1 \;\cdots\; u_n\,]$ (a row, without commas); the vector $(3, -1)$ is a column.
- **Pictures in $\mathbb{R}^2$ and $\mathbb{R}^3$.** A vector is a point, or an arrow from $\mathbf{0}$ to that point. Adding $\mathbf{v}$ to $\mathbf{u}$ moves each entry of $\mathbf{u}$ by the entries of $\mathbf{v}$, so the arrow from $\mathbf{u}$ to $\mathbf{u} + \mathbf{v}$ is a copy of the arrow from $\mathbf{0}$ to $\mathbf{v}$. Therefore $\mathbf{0}$, $\mathbf{u}$, $\mathbf{u} + \mathbf{v}$, $\mathbf{v}$ are the corners of a parallelogram (the **parallelogram rule**). The multiple $c\mathbf{u}$ lies on the line through $\mathbf{0}$ and $\mathbf{u}$, $|c|$ times as far from $\mathbf{0}$, on the opposite side when $c < 0$.
- **Algebra rules.** For $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in $\mathbb{R}^n$ and scalars c, d:

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}, \qquad (\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w}), \qquad \mathbf{u} + \mathbf{0} = \mathbf{u}, \qquad \mathbf{u} + (-\mathbf{u}) = \mathbf{0},$$
$$c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}, \qquad (c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}, \qquad c(d\mathbf{u}) = (cd)\mathbf{u}, \qquad 1\mathbf{u} = \mathbf{u}.$$

  Each holds because the same rule holds for numbers, applied entry by entry.

Definition. Given vectors $\mathbf{v}_1, \ldots, \mathbf{v}_p$ in $\mathbb{R}^n$ and scalars $c_1, \ldots, c_p$, the vector $c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p$ is a **linear combination** of $\mathbf{v}_1, \ldots, \mathbf{v}_p$ with **weights** $c_1, \ldots, c_p$. The set of all linear combinations of $\mathbf{v}_1, \ldots, \mathbf{v}_p$ is $\mathrm{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$, the subset of $\mathbb{R}^n$ **spanned** by $\mathbf{v}_1, \ldots, \mathbf{v}_p$.

Theorem (A vector equation is a linear system (Lay 1.3))

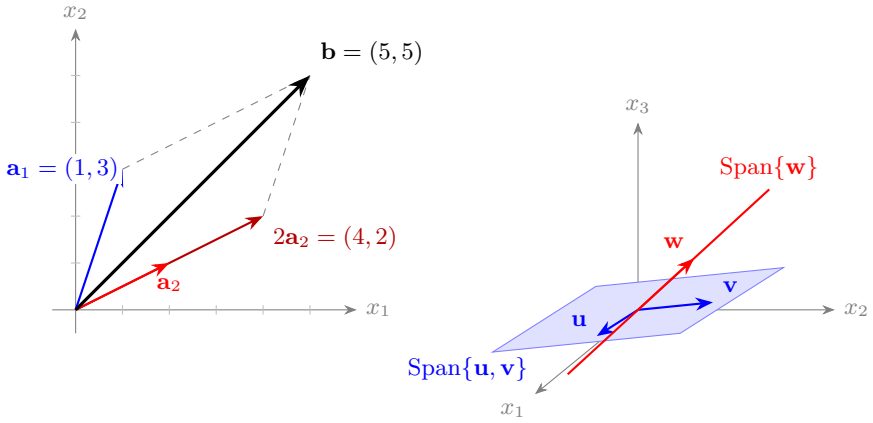
The vector equation $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$ has the same solution set as the linear system whose augmented matrix is $[\,\mathbf{a}_1 \;\; \mathbf{a}_2 \;\; \cdots \;\; \mathbf{a}_n \;\; \mathbf{b}\,]$. In particular, $\mathbf{b}$ is in $\mathrm{Span}\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$ exactly when that system is consistent.

Proof. 1. Write a_{ij} for entry i of $\mathbf{a}_j$. By the definitions of sum and scalar multiple, entry i of $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ is $a_{i1}x_1 + \cdots + a_{in}x_n$.

2. Two vectors are equal exactly when each entry is equal. So the vector equation holds exactly when $a_{i1}x_1 + \cdots + a_{in}x_n = b_i$ for $i = 1, \ldots, m$. This is the linear system whose augmented matrix has $\mathbf{a}_j$ as column j and $\mathbf{b}$ as the last column.

3. $\mathbf{b}$ is in the span exactly when some weights $x_1, \ldots, x_n$ satisfy the vector equation, that is, when the system has a solution.

□



Left: the question's vector equation; $\mathbf{b} = \mathbf{a}_1 + 2\mathbf{a}_2$ is the fourth corner of the parallelogram with sides $\mathbf{a}_1$ and $2\mathbf{a}_2$. Right: in $\mathbb{R}^3$, the span of one nonzero vector is a line through $\mathbf{0}$, and the span of two vectors that are not multiples of each other is a plane through $\mathbf{0}$.

The question answered. By the theorem, the two equations say the same as one vector equation:

$$x_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix},$$

that is, "$(5, 5)$ is the linear combination of $(1, 3)$ and $(2, 1)$ with weights x_1, x_2". Here $(1, 3) + 2(2, 1) = (5, 5)$, so $(x_1, x_2) = (1, 2)$ (left panel of the figure).

Worked example: membership in a span. Let

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \quad \mathbf{a}_3 = \begin{bmatrix} 2 \\ -5 \\ 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 7 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 4 \\ -9 \\ 14 \end{bmatrix}.$$

Is $\mathbf{b}$ in $\mathrm{Span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$? Is $\mathbf{c}$? Apply the operations $R_2 + 2R_1$, $R_3 - 3R_1$, then $R_3 - 2R_2$ to both augmented matrices:

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ -2 & 1 & -5 & 0 \\ 3 & 2 & 4 & 7 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right], \qquad \left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ -2 & 1 & -5 & -9 \\ 3 & 2 & 4 & 14 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 4 \end{array}\right].$$

1. For $\mathbf{b}$: no row $[\,0\;0\;0\;\;b\,]$ with $b \neq 0$, so the system is consistent and $\mathbf{b}$ is in the span. Taking the free variable $x_3 = 0$ gives $x_2 = 2$ and $x_1 = 1$, so $\mathbf{b} = \mathbf{a}_1 + 2\mathbf{a}_2$. Check: $(1, -2, 3) + (0, 2, 4) = (1, 0, 7)$.

2. For $\mathbf{c}$: the last row is the equation $0 = 4$, so $\mathbf{c}$ is not in the span.

Pictures of a span in $\mathbb{R}^3$. $\text{Span}\{\mathbf{v}\} = \{c\mathbf{v}\}$ for $\mathbf{v} \neq \mathbf{0}$ is the line through $\mathbf{0}$ and $\mathbf{v}$. $\text{Span}\{\mathbf{u}, \mathbf{v}\}$, when neither vector is a multiple of the other, is the plane through $\mathbf{0}$ containing $\mathbf{u}$ and $\mathbf{v}$ (right panel). Every span contains $\mathbf{0}$ (all weights 0). In the worked example $\mathbf{a}_3 = 2\mathbf{a}_1 - \mathbf{a}_2$, so the three vectors span only the plane of $\mathbf{a}_1$ and $\mathbf{a}_2$, which contains $\mathbf{b}$ but not $\mathbf{c}$.

Shortcuts

- **"Is $\mathbf{b}$ in $\text{Span}\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$?"** means "is the system with augmented matrix $[\,\mathbf{a}_1 \cdots \mathbf{a}_n \;\; \mathbf{b}\,]$ consistent?": one row reduction. Reason: a vector equation is a linear system (Lay 1.3). Instance: the worked example.
- **Free memberships**: $\mathbf{0}$ and each $\mathbf{a}_j$ are in $\text{Span}\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$, with all weights 0, or weight 1 on $\mathbf{a}_j$ and 0 elsewhere. Instance: $\mathbf{a}_2 = 0\mathbf{a}_1 + 1\mathbf{a}_2 + 0\mathbf{a}_3$.
- **Drop a redundant vector**: if one vector is a combination of the others, removing it does not change the span. Reason, in the worked example: with $\mathbf{a}_3 = 2\mathbf{a}_1 - \mathbf{a}_2$, every combination $c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + c_3\mathbf{a}_3$ equals $(c_1 + 2c_3)\mathbf{a}_1 + (c_2 - c_3)\mathbf{a}_2$, so it already lies in $\text{Span}\{\mathbf{a}_1, \mathbf{a}_2\}$.

Traps

- $\text{Span}\{\mathbf{u}, \mathbf{v}\}$ is a plane only if neither vector is a multiple of the other. $\text{Span}\{(1, 2, 0), (2, 4, 0)\}$ is a line, because $(2, 4, 0) = 2(1, 2, 0)$.
- A span always contains $\mathbf{0}$, so a line or plane that misses $\mathbf{0}$ is not a span. The line $x_2 = x_1 + 1$ in $\mathbb{R}^2$ misses $(0, 0)$, so it is not the span of any vectors.

§2.2 The matrix equation $A\mathbf{x} = \mathbf{b}$ (Lay 1.4)

The question. For the matrix

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix},$$

the equation $A\mathbf{x} = \mathbf{b}$ has a solution for $\mathbf{b} = (1, 2)$ but not for $\mathbf{b} = (1, 0)$. For which $\mathbf{b}$ does it have a solution? Which matrices make $A\mathbf{x} = \mathbf{b}$ solvable for *every* $\mathbf{b}$, and how can we tell without trying every $\mathbf{b}$?

Definition. If $A = [\,\mathbf{a}_1 \;\cdots\; \mathbf{a}_n\,]$ is an $m \times n$ matrix with columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$ and $\mathbf{x}$ is in $\mathbb{R}^n$, then

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

$A\mathbf{x}$ is defined only when the number of columns of A equals the number of entries of $\mathbf{x}$.

- **Row-vector rule**: entry i of $A\mathbf{x}$ is $a_{i1}x_1 + \cdots + a_{in}x_n$, the sum of the products of the entries of row i of A with the entries of $\mathbf{x}$. Reason: entry i of $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ is $x_1a_{i1} + \cdots + x_na_{in}$. For example

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 4 + 6 - 7 \\ 0 - 15 + 21 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}.$$

- The **identity matrix** I_n has 1's on the diagonal and 0 elsewhere. $I_n\mathbf{x} = \mathbf{x}$, because by the row-vector rule entry i of $I_n\mathbf{x}$ is $1 \cdot x_i$.

Worked example: three ways to write a system. The system $x_1 + 2x_2 - x_3 = 4$, $-5x_2 + 3x_3 = 1$ is the vector equation and the matrix equation

$$x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ -5 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}.$$

Column j of the coefficient matrix holds the coefficients of x_j.

Theorem (Three ways to write one system (Lay 1.4, Thm. 3))

For an $m \times n$ matrix A with columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$ and $\mathbf{b}$ in $\mathbb{R}^m$, the matrix equation $A\mathbf{x} = \mathbf{b}$, the vector equation $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$, and the linear system with augmented matrix $[\,A \;\; \mathbf{b}\,]$ all have the same solution set. So $A\mathbf{x} = \mathbf{b}$ has a solution exactly when $\mathbf{b}$ is a linear combination of the columns of A.

Proof. 1. By the definition of $A\mathbf{x}$, the matrix equation and the vector equation are the same equation.

2. By a vector equation is a linear system (Lay 1.3), the vector equation has the same solution set as the system with augmented matrix $[\,\mathbf{a}_1 \cdots \mathbf{a}_n \;\; \mathbf{b}\,] = [\,A \;\; \mathbf{b}\,]$, and it has a solution exactly when $\mathbf{b}$ is in the span of the columns.

□

Worked example: the set of all b for which $A\mathbf{x} = \mathbf{b}$ is consistent. Let

$$A = \begin{bmatrix} 1 & 3 & 4 \\ -4 & 2 & -6 \\ -3 & -2 & -7 \end{bmatrix}$$

and keep $\mathbf{b} = (b_1, b_2, b_3)$ as letters. Row reduce $[\,A \;\; \mathbf{b}\,]$:

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & b_1 \\ -4 & 2 & -6 & b_2 \\ -3 & -2 & -7 & b_3 \end{array}\right] \xrightarrow[R_3+3R_1]{R_2+4R_1} \left[\begin{array}{ccc|c} 1 & 3 & 4 & b_1 \\ 0 & 14 & 10 & b_2+4b_1 \\ 0 & 7 & 5 & b_3+3b_1 \end{array}\right] \xrightarrow{R_3-\frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 3 & 4 & b_1 \\ 0 & 14 & 10 & b_2+4b_1 \\ 0 & 0 & 0 & b_1-\frac{1}{2}b_2+b_3 \end{array}\right]$$

1. The last row is the equation $0 = b_1 - \frac{1}{2}b_2 + b_3$. So $A\mathbf{x} = \mathbf{b}$ is consistent exactly when $b_1 - \frac{1}{2}b_2 + b_3 = 0$, that is, $2b_1 - b_2 + 2b_3 = 0$.
2. This set of $\mathbf{b}$ is a plane through $\mathbf{0}$ in $\mathbb{R}^3$: it is Span of the columns of A. Check: each column satisfies the equation, since $2 + 4 - 6 = 0$, $6 - 2 - 4 = 0$, $8 + 6 - 14 = 0$.
3. For example $\mathbf{b} = (1, 2, 0)$ gives $2 - 2 + 0 = 0$: consistent. $\mathbf{b} = (1, 0, 0)$ gives $2 \neq 0$: inconsistent.

Theorem (When $A\mathbf{x} = \mathbf{b}$ is solvable for every $\mathbf{b}$ (Lay 1.4, Thm. 4))

For an $m \times n$ matrix A the following are equivalent:

(a) for each $\mathbf{b}$ in $\mathbb{R}^m$, $A\mathbf{x} = \mathbf{b}$ has a solution;

(b) each $\mathbf{b}$ in $\mathbb{R}^m$ is a linear combination of the columns of A;

(c) the columns of A span $\mathbb{R}^m$;

(d) A has a pivot position in every row.

The idea

Statements (a), (b), (c) are restatements of each other by Theorem 3 and the definition of span. The content is (a) $\iff$ (d): a condition on infinitely many right-hand sides becomes one row reduction of A. To find a bad $\mathbf{b}$ when a row has no pivot, start from the false equation $0 = 1$ in that row and work backward.

Proof. 1. (a) $\iff$ (b) by Theorem 3, and (b) $\iff$ (c) by the definition of Span.

2. (d)⇒(a). Let U be an echelon form of A, reached by some row operations. Apply the same operations to $[\,A \;\; \mathbf{b}\,]$; the result is $[\,U \;\; \mathbf{d}\,]$ for some $\mathbf{d}$. Every row of U has a pivot, so every row of $[\,U \;\; \mathbf{d}\,]$ has its leading entry in U, and no row is $[\,0 \cdots 0 \; b\,]$ with $b \neq 0$. By (W2) the system is consistent.

3. (not d)⇒(not a). Suppose some row of A has no pivot. Then A has fewer than m pivots. Each nonzero row of U has a leading entry, which is a pivot (W4), so U has fewer than m nonzero rows and therefore a zero row; zero rows are at the bottom, so row m of U is zero.

4. Let $\mathbf{d} = (0, \ldots, 0, 1)$. The last row of $[\,U \;\; \mathbf{d}\,]$ is $[\,0 \cdots 0 \; 1\,]$, so this system is inconsistent. Apply to $[\,U \;\; \mathbf{d}\,]$ the operations that undo the ones taking A to U (W1). The result is $[\,A \;\; \mathbf{b}\,]$ for some $\mathbf{b}$, with the same empty solution set. So $A\mathbf{x} = \mathbf{b}$ has no solution for this $\mathbf{b}$.

□

Two consequences.

- If $m > n$ (more rows than columns), A has at most $n < m$ pivots (W3), so some row has no pivot and the columns do not span $\mathbb{R}^m$.
- If A is square ($n \times n$), then a pivot in every column is the same as a pivot in every row. Reason: either condition says there are n pivots, because each row and each column holds at most one pivot (W3).

The question answered. Row reduce with $\mathbf{b}$ as letters:

$$\left[\begin{array}{cc|c} 1 & 3 & b_1 \\ 2 & 6 & b_2 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 3 & b_1 \\ 0 & 0 & b_2 - 2b_1 \end{array}\right].$$

$A\mathbf{x} = \mathbf{b}$ has a solution exactly when $b_2 = 2b_1$: the line through $\mathbf{0}$ and $(1, 2)$, which contains $(1, 2)$ but not $(1, 0)$. Row 2 of A has no pivot, so by Theorem 4 not every $\mathbf{b}$ works. In general, $A\mathbf{x} = \mathbf{b}$ is solvable for every $\mathbf{b}$ exactly when A has a pivot in every row, which one row reduction of A decides.

Theorem (Linearity of $A\mathbf{x}$ (Lay 1.4, Thm. 5))
If A is $m \times n$, $\mathbf{u}, \mathbf{v}$ are in $\mathbb{R}^n$ and c is a scalar, then $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$ and $A(c\mathbf{u}) = c(A\mathbf{u})$.

Proof. 1. The entries of $\mathbf{u} + \mathbf{v}$ are $u_j + v_j$, so by the definition of $A\mathbf{x}$ and the rule $(c + d)\mathbf{w} = c\mathbf{w} + d\mathbf{w}$,

$$A(\mathbf{u} + \mathbf{v}) = (u_1 + v_1)\mathbf{a}_1 + \cdots + (u_n + v_n)\mathbf{a}_n = u_1\mathbf{a}_1 + v_1\mathbf{a}_1 + \cdots + u_n\mathbf{a}_n + v_n\mathbf{a}_n.$$

2. Regrouping the terms (vector addition is commutative and associative),

$$A(\mathbf{u} + \mathbf{v}) = (u_1\mathbf{a}_1 + \cdots + u_n\mathbf{a}_n) + (v_1\mathbf{a}_1 + \cdots + v_n\mathbf{a}_n) = A\mathbf{u} + A\mathbf{v}.$$

3. The entries of $c\mathbf{u}$ are cu_j, so by the rules $(cd)\mathbf{w} = c(d\mathbf{w})$ and $c(\mathbf{v} + \mathbf{w}) = c\mathbf{v} + c\mathbf{w}$,

$$A(c\mathbf{u}) = (cu_1)\mathbf{a}_1 + \cdots + (cu_n)\mathbf{a}_n = c(u_1\mathbf{a}_1) + \cdots + c(u_n\mathbf{a}_n) = c(u_1\mathbf{a}_1 + \cdots + u_n\mathbf{a}_n) = c(A\mathbf{u}).$$

□

Shortcuts

- **Row-vector rule** for computing $A\mathbf{x}$ by hand: one sum of products per entry. Reason and instance: above.
- **"Do the columns span $\mathbb{R}^m$?"** means "is there a pivot in every row of A?": reduce A alone, with no right-hand side. Reason: Theorem 4. Instance: the question, where row 2 has no pivot.
- **"For which $\mathbf{b}$ is $A\mathbf{x} = \mathbf{b}$ consistent?"** Reduce $[\,A \;\; \mathbf{b}\,]$ with the entries of $\mathbf{b}$ as letters; each zero row of the echelon form of A gives one equation that $\mathbf{b}$ must satisfy. Instance: the worked example and the question.
- **Size check**: if A has more rows than columns, its columns do not span $\mathbb{R}^m$. Reason: it has at most $n < m$ pivots (W3), so some row has no pivot (Theorem 4). Instance: a 3×2 matrix has at most 2 pivots for 3 rows.
- **Square matrices**: a pivot in every column gives a pivot in every row, so the columns span $\mathbb{R}^n$. Reason: either condition says there are n pivots, because each row and each column holds at most one pivot; then Theorem 4. Instance: for the coefficient matrix of the 1.3 question,

$$\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \xrightarrow{R_2 - 3R_1} \begin{bmatrix} 1 & 2 \\ 0 & -5 \end{bmatrix},$$

  so both columns have pivots, both rows have pivots, and $(1, 3)$, $(2, 1)$ span $\mathbb{R}^2$.

Traps

- $A\mathbf{x}$ needs (number of columns of A) = (number of entries of $\mathbf{x}$): a 2×3 matrix times a vector in $\mathbb{R}^2$ is not defined.
- "Pivot in every row" is about A, not $[\,A \;\; \mathbf{b}\,]$. In the question with $\mathbf{b} = (1, 0)$, the augmented matrix reduces to the matrix below on the left, which has a pivot in every row, yet the system has no solution. And a pivot in every row of $[\,A \;\; \mathbf{b}\,]$ does not mean inconsistency either: the matrix below on the right has one and its system has the solution $(3, 4)$.

$$\left[\begin{array}{cc|c} 1 & 3 & 1 \\ 0 & 0 & -2 \end{array}\right] \qquad \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 4 \end{array}\right]$$

- Consistency for one $\mathbf{b}$ says nothing about every $\mathbf{b}$: in the question, $A\mathbf{x} = (1, 2)$ is consistent, but the columns of A do not span $\mathbb{R}^2$.

§2.3 Solution sets of linear systems (Lay 1.5)

The question. For the matrix below, both $W\mathbf{x} = \mathbf{0}$ and $W\mathbf{x} = (2, 5, 3)$ have infinitely many solutions. What do the two solution sets look like, and how are they related?

$$W = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 7 & 4 \\ 1 & 4 & 3 \end{bmatrix}$$

Definition. A system $A\mathbf{x} = \mathbf{0}$ is **homogeneous**. It always has the **trivial solution** $\mathbf{x} = \mathbf{0}$. A solution $\mathbf{x} \neq \mathbf{0}$ is **nontrivial**: at least one entry is nonzero.

- $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution *if and only if* it has at least one free variable. Reason: it is consistent (it has $\mathbf{0}$), so by (W2) its solution is unique, namely $\mathbf{0}$, exactly when there is no free variable.
- If there is no free variable, the solution set is $\{\mathbf{0}\}$.

Worked example: a homogeneous system. Solve $W\mathbf{x} = \mathbf{0}$. The last column of $[\,W \;\; \mathbf{0}\,]$ stays zero under every row operation, so reduce W alone:

$$\begin{bmatrix} 1 & 3 & 1 \\ 2 & 7 & 4 \\ 1 & 4 & 3 \end{bmatrix} \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 - 3R_2} \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

1. x_3 is free; the rows say $x_1 - 5x_3 = 0$ and $x_2 + 2x_3 = 0$, so $x_1 = 5x_3$ and $x_2 = -2x_3$.
2. Write the solution as a vector and factor out the free variable:

$$\mathbf{x} = \begin{bmatrix} 5x_3 \\ -2x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}.$$

3. The solution set is $\mathrm{Span}\{(5, -2, 1)\}$, a line through $\mathbf{0}$.

Definition. Writing the solutions as $\mathbf{x} = \mathbf{p} + t_1\mathbf{v}_1 + \cdots + t_k\mathbf{v}_k$, with the free variables $t_1, \ldots, t_k$ as weights, is the **parametric vector form** of the solution set. It is an **explicit** description (it lists the solutions); the equations themselves are an **implicit** description (they give a test a vector must pass).

In general the solution set of $A\mathbf{x} = \mathbf{0}$ is $\mathrm{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$, with one vector for each free variable.

Worked example: a nonhomogeneous system. Now solve $W\mathbf{x} = (2, 5, 3)$ with the same row operations:

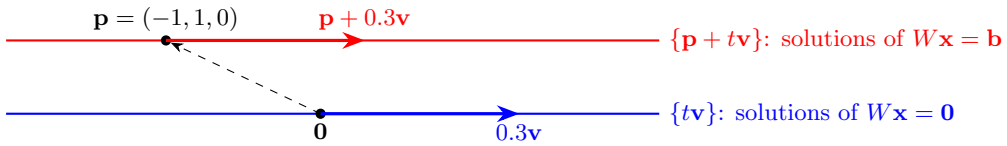
$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 2 \\ 2 & 7 & 4 & 5 \\ 1 & 4 & 3 & 3 \end{array}\right] \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 3 & 1 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & 1 \end{array}\right] \xrightarrow{R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 3 & 1 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 - 3R_2} \left[\begin{array}{ccc|c} 1 & 0 & -5 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

1. $x_1 = -1 + 5x_3$ and $x_2 = 1 - 2x_3$, with x_3 free.
2. Split into a constant vector and a multiple of x_3:

$$\mathbf{x} = \begin{bmatrix} -1 + 5x_3 \\ 1 - 2x_3 \\ x_3 \end{bmatrix} = \underbrace{\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}}_{\mathbf{p}} + x_3 \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}.$$

3. The vector $(5, -2, 1)$ is the one from the homogeneous system, and $\mathbf{p}$ is the solution with $x_3 = 0$.

Lines in parametric form. The set of all $\mathbf{x} = \mathbf{p} + t\mathbf{v}$ (t real, $\mathbf{v} \neq \mathbf{0}$) is the line through $\mathbf{p}$ parallel to $\mathbf{v}$: it is the line $\mathrm{Span}\{\mathbf{v}\}$ moved so that $\mathbf{0}$ goes to $\mathbf{p}$. The line through two points $\mathbf{p} \neq \mathbf{q}$ is $\mathbf{x} = \mathbf{p} + t(\mathbf{q} - \mathbf{p})$: its direction is $\mathbf{q} - \mathbf{p}$, and $t = 0$ gives $\mathbf{p}$, $t = 1$ gives $\mathbf{q}$.



The question's two solution sets, drawn to scale in the plane of $\mathbb{R}^3$ that contains both lines (it passes through $\mathbf{0}$, $\mathbf{v} = (5, -2, 1)$ and $\mathbf{p}$). The solutions of $W\mathbf{x} = \mathbf{b}$ are the solutions of $W\mathbf{x} = \mathbf{0}$ moved by the particular solution $\mathbf{p}$: two parallel lines.

Theorem (Solutions of $A\mathbf{x} = \mathbf{b}$ (Lay 1.5, Thm. 6))
Suppose $A\mathbf{x} = \mathbf{b}$ is consistent and $\mathbf{p}$ is one solution. Then the solution set of $A\mathbf{x} = \mathbf{b}$ is the set of all vectors $\mathbf{w} = \mathbf{p} + \mathbf{v}_h$, where $\mathbf{v}_h$ is any solution of $A\mathbf{x} = \mathbf{0}$.

Proof. 1. If $A\mathbf{v}_h = \mathbf{0}$, then by linearity (Theorem 5)

$$A(\mathbf{p} + \mathbf{v}_h) = A\mathbf{p} + A\mathbf{v}_h = \mathbf{b} + \mathbf{0} = \mathbf{b}.$$

So every $\mathbf{p} + \mathbf{v}_h$ is a solution.

2. If $A\mathbf{w} = \mathbf{b}$, let $\mathbf{v}_h = \mathbf{w} - \mathbf{p}$. By linearity, $A\mathbf{v}_h = A\mathbf{w} - A\mathbf{p} = \mathbf{b} - \mathbf{b} = \mathbf{0}$, and $\mathbf{w} = \mathbf{p} + \mathbf{v}_h$. So every solution has this form.

□

Consequence. A consistent system $A\mathbf{x} = \mathbf{b}$ has exactly one solution if and only if $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. Reason: by the theorem, the solutions are the vectors $\mathbf{p} + \mathbf{v}_h$, and there is only one of them exactly when $\mathbf{v}_h = \mathbf{0}$ is the only choice.

Writing a solution set in parametric vector form.

1. Row reduce $[\,A \;\; \mathbf{b}\,]$ to reduced echelon form; stop if a row $[\,0 \cdots 0 \; b\,]$ with $b \neq 0$ appears.
2. Express each basic variable in terms of the free variables.
3. Write $\mathbf{x}$ as a vector whose entries depend only on the free variables.
4. Split $\mathbf{x}$ into a constant vector plus one vector for each free variable, with that free variable as its weight.

The question answered. $W\mathbf{x} = \mathbf{0}$ has the line $\mathrm{Span}\{(5, -2, 1)\}$ through $\mathbf{0}$ as its solution set. $W\mathbf{x} = (2, 5, 3)$ has the parallel line through $\mathbf{p} = (-1, 1, 0)$. The theorem says this is always the relation: the solutions of $A\mathbf{x} = \mathbf{b}$ are those of $A\mathbf{x} = \mathbf{0}$ moved by one particular solution.

Shortcuts

- **Nontrivial solutions of $A\mathbf{x} = \mathbf{0}$** exist exactly when there is a free variable, and with more columns than rows there always is one. Reason: (W3) gives at most m pivots for $n > m$ columns. Instance: a 2×4 matrix has at most 2 pivots, so at least 2 free variables.
- **Homogeneous systems**: reduce A, not $[\,A \;\; \mathbf{0}\,]$. Reason: row operations keep a zero column zero. Instance: the reduction of W above.
- **Check a parametric vector form** by multiplying: the constant vector must satisfy $A\mathbf{p} = \mathbf{b}$ and each other vector $A\mathbf{v}_i = \mathbf{0}$. Reason: Theorem 6. Instance: $W(-1, 1, 0) = (2, 5, 3)$ and $W(5, -2, 1) = \mathbf{0}$.
- **One equation in $\mathbb{R}^3$**, $ax_1 + bx_2 + cx_3 = d$ with $(a, b, c) \neq \mathbf{0}$, has one basic variable and two free ones, so its solution set is a plane: through $\mathbf{0}$ if $d = 0$, otherwise parallel to that plane. Instance: $x_1 - x_2 + 2x_3 = 3$ gives

$$\mathbf{x} = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}.$$

Traps

- The solution set of $A\mathbf{x} = \mathbf{b}$ with $\mathbf{b} \neq \mathbf{0}$ is *not* a span: it does not contain $\mathbf{0}$, because $A\mathbf{0} = \mathbf{0} \neq \mathbf{b}$. The solution line of $W\mathbf{x} = (2, 5, 3)$ misses $\mathbf{0}$.
- The particular solution is not unique; any solution works. With $\mathbf{p}' = \mathbf{p} + (5, -2, 1) = (4, -1, 1)$, the vectors $\mathbf{p}' + x_3(5, -2, 1)$ form the same line.
- Every entry of $\mathbf{x}$ must be written in terms of the free variables only. Writing the solutions of $W\mathbf{x} = \mathbf{0}$ as $(5x_3,\, x_2,\, x_3)$ leaves the basic variable x_2 in place, and this describes a plane, not the solution line: $(5, 0, 1)$ has that form, but $W(5, 0, 1) = (6, 14, 8) \neq \mathbf{0}$.
- A nontrivial solution may have zero entries; it needs only one nonzero entry. For the single equation $x_1 = 0$ in $\mathbb{R}^2$, the vector $(0, 1)$ is a nontrivial solution.

§3 Homework this week

HW1, Sections 1.3–1.4, and HW2, Section 1.5.

Section 1.3.

- **#6, #10** (vector equation ↔ system). Each entry gives one scalar equation; the coefficients of each variable form one column (theorem in 1.3).
- **#7** (vectors on a grid). Count steps along $\mathbf{u}$ and $\mathbf{v}$ from $\mathbf{0}$; for "every vector", ask whether $\mathbf{u}$ and $\mathbf{v}$ are multiples of each other.
- **#13, #17** (is $\mathbf{b}$ a combination; which h). Row reduce the augmented matrix with $\mathbf{b}$ as last column and look for a row $[\,0 \cdots 0 \; b\,]$, $b \neq 0$ (worked example, P2).
- **#19** (describe a span). Check whether one vector is a multiple of the other (pictures of a span).
- **#22** (columns that miss some $\mathbf{b}$). Make one column a combination of the others, then choose $\mathbf{b}$ off their plane and verify by row reduction.
- **#24** (lists of five numbers). The definition of $\mathbb{R}^5$.

Section 1.4.

- **#1, #2** (products). Check sizes first; then use the definition and the row-vector rule (P4).
- **#9** (system to vector and matrix equation). Column j of A holds the coefficients of x_j (worked example).
- **#11** (solve $A\mathbf{x} = \mathbf{b}$). Row reduce $[\,A \;\; \mathbf{b}\,]$ (Theorem 3) and write $\mathbf{x}$ as a vector.
- **#13, #14** (is $\mathbf{u}$ in the span of the columns?). Is $A\mathbf{x} = \mathbf{u}$ consistent?
- **#15** (which $\mathbf{b}$ work). Reduce $[\,A \;\; \mathbf{b}\,]$ with b_1, b_2 as letters (worked example on the set of $\mathbf{b}$).

- **#19** (do the columns span $\mathbb{R}^4$?). Theorem 4: a pivot in every row of A.
- **#32** (solution sets agree). Theorem 3.
- **#34** (pivot in every row of $[\,A\ \ \mathbf{b}\,]$). Where can the pivot of the last row be? Existence and uniqueness theorem (Week 1).

Section 1.5.

- **#1** (nontrivial solutions?). Reduce only far enough to count free variables.
- **#6, #11** (parametric vector form of $A\mathbf{x} = \mathbf{0}$). Reduced echelon form, then factor out each free variable (method box).
- **#15** (check solutions). Multiply A by each vector of your parametric form (shortcut "check a parametric vector form").
- **#18** (a line in $\mathbb{R}^4$). Write $\mathbf{x} = \mathbf{p} + x_4\mathbf{v}$ by separating the constant part from the free-variable part.
- **#24** (line through $\mathbf{a}$ parallel to $\mathbf{b}$). $\mathbf{x} = \mathbf{a} + t\mathbf{b}$ (lines in parametric form).
- **#27** (homogeneous systems consistent?). Definition of a homogeneous system: try $\mathbf{x} = \mathbf{0}$.
- **#28** (entries of a nontrivial solution). Definition of nontrivial: how many entries must be nonzero?
- **#30** ($\mathbf{x} = x_2\mathbf{u} + x_3\mathbf{v}$ a plane?). Pictures of a span in $\mathbb{R}^3$: what does the span of two vectors look like in each case?
- **#38** (unique solution of $A\mathbf{x} = \mathbf{b}$ versus $A\mathbf{x} = \mathbf{0}$). The consequence of Theorem 6.

§4 Problem set

P1. Write $\mathbf{b} = (5, 1, 12)$ as a linear combination of $\mathbf{a}_1 = (1, 0, 2)$, $\mathbf{a}_2 = (2, 1, 3)$, $\mathbf{a}_3 = (1, -1, 4)$, or show that this is impossible.

P2. For which h is $\mathbf{y} = (3, h, 1)$ in the plane spanned by $\mathbf{v}_1 = (1, 2, -1)$ and $\mathbf{v}_2 = (2, 5, 0)$? For that h, write $\mathbf{y}$ as a combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.

P3. For each matrix, decide whether its columns span $\mathbb{R}^3$. For the one whose columns do not, find the set of all $\mathbf{b}$ for which $A\mathbf{x} = \mathbf{b}$ is consistent.

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 3 & 0 \\ 1 & 3 & -3 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 4 & 1 \end{bmatrix}.$$

P4. Compute the product below by the definition (combination of columns) and by the row-vector rule:

$$\begin{bmatrix} 2 & -1 \\ 0 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ -2 \end{bmatrix}.$$

P5. Write the solution set of $A\mathbf{x} = \mathbf{0}$ in parametric vector form, for

$$A = \begin{bmatrix} 1 & 3 & -3 & 7 \\ 0 & 1 & -4 & 5 \\ 2 & 7 & -10 & 19 \end{bmatrix}.$$

P6. Write the solution set of $x_1 + 2x_2 - x_3 = 1$, $2x_1 + 5x_2 + x_3 = 4$, $3x_1 + 7x_2 = 5$ in parametric vector form, and describe it geometrically.

P7. True or false, with a reason. (a) If A is 3×2, the columns of A can span $\mathbb{R}^3$. (b) If the columns of a 4×4 matrix A span $\mathbb{R}^4$, then $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. (c) If A is 3×5, then $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution.

P8. Describe the solution set of the single equation $x_1 + 2x_2 - 3x_3 = 4$ in parametric vector form, and compare it with the solution set of $x_1 + 2x_2 - 3x_3 = 0$.